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United States Patent Application Publication

20250286302

Kind Code

A1

Publication Date

September 11, 2025

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CONNECTOR AND CONNECTOR MANUFACTURING METHOD

Abstract

A connector **10** comprises an electric wire **40** having a core wire **41** and an insulating coating **42** covering the outer periphery of the core wire **41**, a terminal **50** connected to the core wire **41**, a connector housing **20** having a terminal storage part **21** in which the terminal **50** is stored, and a rear holder **30** attached to the rear end portion of the connector housing **20**. The rear holder **30** has an electric wire insertion hole **33** in which the electric wire **40** is inserted, the terminal **50** includes an insertion cylindrical part **55** in which the core wire **41** is inserted from the rear and a contact part **56** that is disposed at the rear end portion of the insertion cylindrical part **55** and faces the electric wire insertion hole **33**, the insulating coating **42** of the electric wire **40** has a sheath portion **43** and a body portion **44** as separate portions, and the sheath portion **43** is disposed in front of the body portion **44** and is slidable with respect to the core wire **41**.

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Family ID: 88730366

Appl. No.: 18/862554

Filed (or PCT Filed): April 24, 2023

PCT No.: PCT/JP2023/016076

Foreign Application Priority Data

JP 2022-077959

May. 11, 2022

Publication Classification

Int. Cl.: H01R13/424 (20060101); H01R43/20 (20060101)

U.S. Cl.:

CPC H01R13/424 (20130101); H01R43/20 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a connector and a connector manufacturing method.

BACKGROUND

[0002] Conventionally, a female terminal described in Japanese Patent Laid-Open Publication No. 2019-145208 (Patent Document 1 below) is known. This female terminal is provided with a terminal body including deformable upper and lower connection pieces extending in an extension direction, and a slide portion movable in the extension direction with respect to the terminal body. By moving the slide portion with a core wire of a wire disposed between the upper and lower connection pieces, upper and lower contact portions provided in the slide portion press the upper and lower connection pieces against the core wire, whereby the terminal body and the wire can be electrically connected. Further, a connector provided with a connector housing for accommodating the female terminal and a rear holder to be attached to a rear end part in the extension direction of the connector housing is described as a connector provided with this female terminal in Patent Document 1 below.

PRIOR ART DOCUMENT

Patent Document

[0003] Patent Document 1: JP 2019-145208 A

SUMMARY OF THE INVENTION

Problems to be Solved

[0004] The above connector is manufactured as followed. First, the female terminal is accommodated into the connector housing and the rear holder is assembled with the rear end part of the connector housing. In this state, the rear holder is held at a temporary locking position with respect to the connector housing. Then, the wire is inserted through an insertion hole of the rear holder and the core wire of the wire is inserted into the inside of the female terminal. At this time, the core wire is disposed between the upper and lower connection pieces. Subsequently, by moving the slide portion forward with respect to the terminal body, the upper and lower contact portions press the upper and lower connection pieces against the core wire, and the female terminal and the wire are connected. Finally, the rear holder is moved to a complete locking position forward of the temporary locking position and the female terminal is retained in the connector housing, whereby the manufacturing of the connector is completed.

[0005] In the above connector, a front end part of the core wire may possibly contact a rear end part of the female terminal and it may be difficult to insert the core wire into the inside of the female terminal when the core wire is inserted into the inside of the female terminal.

Means to Solve the Problem

[0006] The present disclosure is directed to a connector with a wire including a core wire and an insulation coating covering an outer periphery of the core wire, a terminal to be connected to the core wire, a connector housing including a terminal accommodating portion for accommodating the terminal inside, and a rear holder to be attached to a rear part of the connector housing, the rear holder including a wire insertion hole, the wire being inserted through the wire insertion hole, the terminal including an insertion tube portion, the core wire being inserted into the insertion tube

portion from behind, and a contact portion disposed in a rear end part of the insertion tube portion and facing the wire insertion hole, the insulation coating of the wire including a sheath portion and a body portion separate from each other, and the sheath portion being disposed forward of the body portion and slidable with respect to the core wire.

Effect of the Invention

[0007] According to the present disclosure, it is possible to provide a connector in which a core wire is easily inserted into the inside of a terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of a connector according to an embodiment.

[0009] FIG. 2 is an exploded perspective view of the connector.

[0010] FIG. 3 is a front view of the connector.

[0011] FIG. 4 is a section along A-A of FIG. 3.

[0012] FIG. 5 is a view showing a state where a wire is inserted in a wire insertion hole and a sheath portion is in contact with a rear end part of a terminal in a cross-section along A-A of FIG. 3.

[0013] FIG. 6 is a view showing a state where the wire is pushed further forward from the state of FIG. 5 and the sheath portion is slid with respect to a core wire in the cross-section along A-A of FIG. 3.

[0014] FIG. 7 is a view showing a state where the wire is pushed further forward from the state of FIG. 6 and the core wire is inside the terminal in the cross-section along A-A of FIG. 3.

[0015] FIG. 8 is a section showing a wire by a two-dot chain line in a cross-section along B-B of FIG. 5.

DETAILED DESCRIPTION TO EXECUTE THE INVENTION

DESCRIPTION OF EMBODIMENTS OF PRESENT DISCLOSURE

[0016] First, embodiments of the present disclosure are listed and described below.

[0017] (1) The connector of the present disclosure is provided with a wire including a core wire and an insulation coating covering an outer periphery of the core wire, a terminal to be connected to the core wire, a connector housing including a terminal accommodating portion for accommodating the terminal inside, and a rear holder to be attached to a rear part of the connector housing, the rear holder including a wire insertion hole, the wire being inserted through the wire insertion hole, the terminal including an insertion tube portion, the core wire being inserted into the insertion tube portion from behind, and a contact portion disposed in a rear end part of the insertion tube portion and facing the wire insertion hole, the insulation coating of the wire including a sheath portion and a body portion separate from each other, and the sheath portion being disposed forward of the body portion and slidable with respect to the core wire.

[0018] According to this configuration, the sheath portion comes into contact with the contact portion by inserting the wire into the wire insertion hole with a front end part of the core wire protected by the sheath portion. With the sheath portion held in contact with the contact portion, the core wire immediately behind the insertion tube portion is substantially in a center of the sheath portion and disposed near a center of the wire insertion hole. Therefore, the core wire is easily inserted into the insertion tube portion.

[0019] (2) Preferably, the insertion tube portion includes a first wall portion and a second wall portion facing the first wall portion, a clearance between the first and second wall portions is set to be smaller than an outer diameter of the wire, and both the first and second wall portions include the contact portion.

[0020] According to this configuration, since both the first and second wall portions include the

contact portion, a state of the sheath portion in contact with the contact portion can be stabilized. In this way, the core wire is more easily inserted into the insertion tube portion.

[0021] (3) Preferably, the rear holder includes a guiding portion projecting more inwardly of the wire insertion hole from an inner wall of the wire insertion hole toward front.

[0022] According to this configuration, the sheath portion can be guided by being caused to slide in contact with the guiding portion. Thus, the core wire is even more easily inserted into the insertion tube portion.

[0023] (4) A connector manufacturing method of the present disclosure is a manufacturing method for a connector with a wire including a core wire and an insulation coating covering an outer periphery of the core wire, a terminal including an insertion tube portion, the core wire being inserted into the insertion tube portion from behind, a connector housing including a terminal accommodating portion, and a rear holder including a wire insertion hole, and includes accommodating the terminal inside the terminal accommodating portion, attaching the rear holder to a rear part of the connector housing with a rear end part of the insertion tube portion at least partially arranged to face the wire insertion hole, forming a sheath portion and a body portion disposed rearward of the sheath portion by cutting off a front end part of the insulation coating of the wire, protecting a front end part of the core wire by the sheath portion by sliding the sheath portion forward with respect to the core wire, inserting the wire having the front end part of the core wire protected by the sheath portion into the wire insertion hole of the rear holder and introducing the core wire into the insertion tube portion while bringing the sheath portion into contact with the rear end part of the insertion tube portion, and connecting the core wire and the terminal.

[0024] Since this connector manufacturing method includes forming the sheath portion in the insulation coating and protecting the front end part of the core wire by the sheath portion, the sheath portion comes into contact with the rear end part of the insertion tube portion if the wire is inserted into the wire insertion hole. With the sheath portion held in contact with the rear end part of the insertion tube portion, the core wire immediately behind the insertion tube portion is substantially in a center of the sheath portion and disposed near a center of the wire insertion hole. Therefore, the core wire is easily inserted into the insertion tube portion.

Details of Embodiment of Present Disclosure

[0025] Hereinafter, an embodiment of the present disclosure is described. The present disclosure is not limited to these illustrations, but is represented by claims and intended to include all changes in the scope of claims and in the meaning and scope of equivalents.

Embodiment

[0026] The embodiment of the present disclosure is described with reference to FIGS. **1** to **8**. In the following description, a direction indicated by an arrow Z is referred to as an upward direction, a direction indicated by an arrow X is referred to as a forward direction and a direction indicated by an arrow Y is referred to as a leftward direction. Note that, for a plurality of identical members, only some members may be denoted by a reference sign and the other members may not be denoted by the reference sign.

Connector

[0027] As shown in FIGS. **1** and **2**, the connector **10** is provided with a connector housing **20**, a rear holder **30**, wires **40** and terminals **50**.

Wire

[0028] The wire **40** is provided with a core wire **41** and an insulation coating **42** made of insulating synthetic resin for surrounding the outer peripheral surface of the core wire **41**. The insulation coating **42** is stripped in an end part of the wire **40** to expose the core wire **41**. The core wire **41** is a so-called single-core wire composed of one metal wire. Note that the core wire **41** may be a stranded wire formed by twisting a plurality of thin metal wires. An arbitrary metal such as copper, copper alloy, aluminum or aluminum alloy can be appropriately selected as a metal constituting the

core wire **41** if necessary. The core wire **41** of this embodiment is made of copper or copper alloy.
Sheath Portion, Body Portion

[0029] The insulation coating **42** is provided with a sheath portion **43** and a body portion **44** separate from the sheath portion **43** and disposed behind the sheath portion **43**. The sheath portion **43** is slidable in an extension direction (front-rear direction) of the wire **40** with respect to the core wire **41**. The sheath portion **43** is formed by being cut off from the body portion **44** by a wire stripper or the like. Although described later, the wire **40** is inserted through a wire insertion hole **33** of the rear holder **30** with a front end part of the core wire **41** protected by the sheath portion **43** (see FIGS. 2 and 5).

Terminal

[0030] The terminal **50** is a female terminal and provided with a terminal body **51** made of metal and a slide portion **52** relatively slidable with respect to the terminal body **51**. The terminal body **51** is formed into a predetermined shape by a known method such as press-working, cutting or casting. An arbitrary metal such as copper, copper alloy, aluminum, aluminum alloy or stainless steel can be appropriately selected as a metal constituting the terminal body **51** if necessary. The terminal body **51** of this embodiment is made of copper or copper alloy. A plating layer may be formed on the surface of the terminal body **51**. An arbitrary metal such as tin, nickel or silver can be appropriately selected as a metal constituting the plating layer if necessary. Tin plating is applied to the terminal body **51** of this embodiment.

Terminal Body

[0031] As shown in FIG. 4, the terminal body **51** includes a connecting tube portion **51A** in the form of a rectangular tube, into which a mating male terminal (not shown) is inserted, a base portion **51B** in the form of a rectangular tube disposed behind the connecting tube portion **51A** and an upper connection piece **51C** and a lower connection piece **51D** extending rearward from the base portion **51B**. The front end of the connecting tube portion **51A** is open to enable the insertion of the mating male terminal.

[0032] A resiliently deformable resilient contact piece **51E** is disposed on a side wall of the connecting tube portion **51A**. The male terminal inserted into the connecting tube portion **51A** contacts the resilient contact piece **51E**. Further, a front end part of the upper wall of the connecting tube portion **51A** is provided with a locking lance **53** made of metal and formed by cutting and raising a part of the connecting tube portion **51A**.

[0033] The upper connection piece **51C** is formed to be resiliently deformable in a vertical direction with a rear end part of the upper wall of the base portion **51B** as a fulcrum. The lower connection piece **51D** is formed to be resiliently deformable in the vertical direction with a rear end part of the lower wall of the base portion **51B** as a fulcrum. The upper and lower connection pieces **51C**, **51D** are formed to have substantially the same length in a front-rear direction.

[0034] The terminal body **51** further includes a terminal window portion **51F** open upward at a position behind the connecting tube portion **51A** and in front of the base portion **51B**. A front end part of the core wire **41** can be detected through the terminal window portion **51F** with the core wire **41** disposed in a space between the upper and lower connection pieces **51C**, **51D**. Detectable means visually confirmable from outside by a worker, detectable from outside by a camera (not shown) or capable of electrically detecting the front end part of the core wire **41** by inserting a probe (not shown) from outside.

Slide Portion

[0035] The slide portion **52** is in the form of a rectangular tube extending in the front-rear direction. The slide portion **52** is formed by a known method such as cutting, casting or press-working if necessary. An arbitrary metal such as copper, copper alloy, aluminum, aluminum alloy or stainless steel can be appropriately selected as a metal constituting the slide portion **52** if necessary. The slide portion **52** of this embodiment is made of copper or copper alloy. A plating layer may be formed on the surface of the slide portion **52**. An arbitrary metal such as tin, nickel or silver can be

appropriately selected as a metal constituting the plating layer if necessary. Tin plating is applied to the slide portion **52** of this embodiment.

[0036] A cross-sectional shape of the slide portion **52** is the same as or somewhat larger than that of a region of the terminal body **51** where the upper and lower connection pieces **51C**, **52D** are provided. In this way, the slide portion **52** is externally fittable to the region of the terminal body **51** where the upper and lower connection pieces **51C**, **52D** are provided.

[0037] An upper contact portion **52A** projecting downward is provided on the lower surface of the upper wall of the slide portion **52**. A lower contact portion **52B** projecting upward is provided on the upper surface of the lower wall of the slide portion **52**.

[0038] The slide portion **52** is slidable between a temporary locking position (position of the slide portion **52** in FIGS. **5** to **7**) and a complete locking position (position of the slide portion **52** in FIG. **4**) set to be forward of the temporary locking position in a state externally fit to the region of the terminal body **51** where the upper and lower connection pieces **51C**, **51D** are provided. With the slide portion **52** held at the complete locking position with respect to the terminal body **51**, the upper contact portion **52A** is in contact with the upper surface of the upper connection piece **51C** from above and the lower contact portion **52B** is in contact with the lower surface of the lower connection piece **51D** from below.

[0039] With the slide portion **52** held at the complete locking position with respect to the terminal body **51**, the upper contact portion **52A** presses the upper connection piece **51C** from above, whereby the upper connection piece **51C** is resiliently deformed downward. Further, the lower contact portion **52B** presses the lower connection piece **51D** from below, whereby the lower connection piece **51D** is resiliently deformed upward. In this way, the core wire **41** is sandwiched in the vertical direction by the resiliently deformed upper and lower connection pieces **51C**, **51D** in a state where the core wire **41** is disposed to extend in the front-rear direction in the space between the upper and lower connection pieces **51C**, **51D** and the slide portion **52** is held at the complete locking position with respect to the terminal body **51**. That is, the upper connection piece **51C** contacts the core wire **41** from above by being pressed downward by the upper contact portion **52A**, and the lower connection piece **51D** contacts the core wire **41** from below by being pressed upward by the lower contact portion **52B**.

[0040] A front part of the upper connection piece **51C** projects downward, a rear part of the lower connection piece **51D** projects upward, and these projecting parts are shifted in the front-rear direction. Thus, the core wire **41** is held in a state bent in the vertical direction. In this way, the core wire **41** is electrically connected to the terminal **50** and firmly held not to come out against the rearward pulling of the wire **40**.

[0041] As shown in FIG. **7**, the front end part of the slide portion **52** is provided with a jig contact portion **54** projecting upward from the upper wall. An unillustrated jig comes into contact with the jig contact portion **54** from behind and the slide portion **52** is pushed forward by the jig, whereby the slide portion **52** is movable forward.

Insertion Tube Portion, First Wall Portion, Second Wall Portion

[0042] As shown in FIGS. **5** to **7**, the slide portion **52** serves as an insertion tube portion **55**, into which the core wire **41** is inserted. As shown in FIG. **8**, the left wall of the insertion tube portion **55** serves as a first wall portion **55A**. The right wall of the insertion tube portion **55** serves as a second wall portion **55B**. The first and second wall portions **55A**, **55B** are facing each other in a direction (lateral direction) orthogonal to the extension direction (front-rear direction) of the wire **40**. The upper end edges of the first and second wall portions **55A**, **55B** are connected by an upper wall. The lower end edges of the first and second wall portions **55A**, **55B** are connected by a lower wall. A clearance **CL1** in the lateral direction between the first and second wall portions **55A**, **55B** is set to be smaller than an outer diameter **D1** of the wire **40** (insulation coating **42**) and larger than a diameter **D2** of the core wire **41**.

Contact Portions

[0043] At least parts of a rear end part of the insertion tube portion **55** serve as contact portions **56** disposed to face the wire insertion hole **33** of the rear holder **30**. That is, the rear end part of the insertion tube portion **55** is at least partially disposed inside the wire insertion hole **33** when viewed from behind. In this embodiment, both the first and second wall portions **55A**, **55B** include the contact portion **56**. Although described later, the contact portions **56** can come into contact with the sheath portion **43** disposed in a tip part of the wire **40** when the wire **40** is inserted into the wire insertion hole **33** (see FIGS. 5 to 7).

Connector Housing

[0044] As shown in FIGS. 2 and 3, the connector housing **20** is made of insulating synthetic resin and includes a terminal accommodating portion **21** for accommodating the plurality of terminals **50**. The connector housing **20** has a substantially rectangular parallelepiped shape flat in the vertical direction and extending in the lateral direction. The terminal accommodating portion **21** includes a plurality of cavities **21A** penetrating therethrough in the front-rear direction. The plurality of cavities **21** are arranged in parallel at intervals in the lateral direction in two upper and lower stages. The cavities **21A** formed in the upper stage and the cavities **21** formed in the lower stage are disposed at positions shifted in the lateral direction when viewed from above. Note that the number of the cavities **21A** is arbitrary and the number of the stages in the vertical direction is also arbitrary.

[0045] The unillustrated male terminal is insertable into the cavity **21A** from front. As shown in FIG. 4, the terminal **50** is inserted into the cavity **21A** from behind and can be accommodated inside the terminal accommodating portion **21**.

[0046] With the terminal **50** accommodated in the cavity **21A**, a connector window portion **21B** is formed at a position corresponding to the terminal window portion **51F** of the terminal **50** in the terminal accommodating portion **21**. By this connector window portion **21B**, the terminal window portion **51F** communicates with outside. The terminal window portion **51F** of the terminal **50** is detectable through the connector window portion **21B**. In this way, the front end part of the core wire **41** is detectable from outside through the connector window portion **21B** and the terminal window portion **51F**.

[0047] As shown in FIG. 2, the connector housing **20** includes a separation wall **22** partitioning between the cavities **21A** in the upper stage and the cavities **21A** in the lower stage. The separation wall **22** extends rearward from rear end parts of the cavities **21A**. Partition walls **23** extending in the front-rear direction and projecting upward or downward are respectively provided on the upper and lower surfaces of the separation wall **22**. The terminals **50** accommodated in the respective cavities **21A** are electrically insulated from the terminals **50** adjacent in the lateral direction by these partition walls **23**.

[0048] Temporary lock portions **24** projecting outward are provided at positions near rear end parts of both left and right side walls of the connector housing **20**, and complete locking portions **25** are provided at positions forward of these temporary locking portions **24**.

[0049] As shown in FIG. 4, a lance hole **26** open upward is provided on a front end side of the cavity **21A** in the upper stage, and a lance hole **26** open downward is provided on a front end side of the cavity **21A** in the lower stage. The locking lance **53** of the terminal **50** is locked to an inner wall constituting the lance hole **26**, whereby the terminal **50** is held in a state accommodated in the cavity **21A**.

Rear Holder

[0050] The rear holder **30** is made of insulating synthetic resin. The rear holder **30** includes a box-shaped receptacle **31** open forward. The receptacle **31** is fit to a rear half of the connector housing **20** from outside. As shown in FIG. 2, lock receiving portions **32** are provided at positions near front end parts of both left and right side walls of the rear holder **30**. The lock receiving portion **32** is substantially gate-shaped.

[0051] By locking the temporary lock portions **24** of the connector housing **20** and the lock

receiving portions **32** of the rear holder **30**, the rear holder **30** is held at a temporary locking position with respect to the connector housing **20**. Further, by locking the complete lock portions **25** of the connector housing **20** and the lock receiving portions **32** of the rear holder **30**, the rear holder **30** is held at a complete locking position with respect to the connector housing **20** (see FIG. 1).

[0052] As shown in FIG. 2, the rear holder **30** includes a plurality of wire insertion holes **33**, through which the wires **40** are inserted. The plurality of wire insertion holes **33** are arranged in parallel in the lateral direction and provided in two upper and lower stages. The respective wire insertion holes **33** are provided at positions corresponding to the respective cavities **21A** of the connector housing **20**. An inner diameter of the wire insertion hole **33** is set to be equal to or somewhat larger than the outer diameter **D1** of the insulation coating **42** of the wire **40**. In this embodiment, as shown in FIG. 8, the inner diameter of the wire insertion hole **33** is larger than the outer diameter **D1** of the insulation coating **42** of the wire **40**. Even if the sheath portion **43** is shifted from a center position of the wire insertion hole **33** and the sheath portion **43** slides in contact with the inner wall of the wire insertion hole **33**, the core wire **41** is disposed inside the insertion tube portion **55** when viewed from behind.

Guiding Portion

[0053] As shown in FIG. 5, the rear holder **30** includes a guiding portion **34** on the inner wall of the wire insertion hole **33**. The guiding portion **34** is disposed on the front end of the wire insertion hole **33**. The guiding portion **34** is formed into a curved surface located more inwardly of the wire insertion hole **33** toward the front. The sheath portion **43** slides in contact with the guiding portion **34**, thereby being guided into the wire insertion hole **33**. In this embodiment, the guiding portion **34** is provided on the inner wall on an upper side of the wire insertion hole **33** in the wire insertion hole **33** in the upper stage, and located lower toward the front. Although not shown, the guiding portion **34** is provided on the inner wall on a lower side of the wire insertion hole **33** in the wire insertion hole **33** in the lower stage, and located higher toward the front.

[0054] In this embodiment, in the terminal **50** accommodated in the terminal accommodating portion **21** in the upper stage, a center position in the vertical direction of the insertion tube portion **55** is shifted downward with respect to a center position in the vertical direction of the wire insertion hole **33** in the upper stage as shown in FIGS. 5 and 8. However, since the rear holder **30** of this embodiment is provided with the guiding portion **34** projecting downward from the inner wall of the wire insertion hole **33** on the front end of the wire insertion hole **33** in the upper stage, the sheath portion **43** can be guided to a position below the center position in the vertical direction of the wire insertion hole **33** by the guiding portion **34**. In this way, the core wire **41** disposed substantially in a center of the sheath portion **43** approaches the center position of the insertion tube portion **55**, wherefore the insertion of the core wire **41** into the insertion tube portion **55** is facilitated.

[0055] Although not shown, the guiding portion **34** projects upward in the wire insertion hole **33** in the lower stage and guides the sheath portion **43** upward. Since the insertion tube portion **55** of the terminal **50** disposed in the terminal accommodating portion **21** in the lower stage is shifted upward with respect to the wire insertion hole **33**, the insertion of the core wire **41** into the insertion tube portion **55** is facilitated by the guiding portion **34**.

[0056] As shown in FIG. 5, a separation wall receiving portion **35** is provided near a vertical center in the back wall of the receptacle **31**. This separation wall receiving portion **35** is open forward and a vertical dimension of an opening is set to be equal to or somewhat larger than a vertical thickness of the separation wall **22** of the connector housing **20**.

[0057] With the rear holder **30** held at the temporary locking position with respect to the connector housing **20**, the separation wall receiving portion **35** is located behind the rear end edge of the separation wall **22** of the connector housing **20**. As shown in FIG. 4, with the rear holder **30** held at the complete locking position with respect to the connector housing **20**, the separation wall **22** of

the connector housing **20** is fit in the opening of the separation wall receiving portion **35** of the rear holder **30**. In this way, a position shift of the rear holder **30** in the vertical direction with respect to the connector housing **20** is suppressed.

[0058] A step **36** is formed at the front end position of the wire insertion hole **33** on the inner wall of the receptacle **31**. This step **36** comes into contact with a rear end part of the slide portion **52** from behind when the rear holder **30** is moved from the temporary locking position to the complete locking position with respect to the connector housing **20**. In this way, the terminal **50** is held in a state accommodated in the cavity **21A** of the connector housing **20** and prevented from coming out rearward.

Connector Manufacturing Method

[0059] Next, an example of a manufacturing method of the connector **10** of this embodiment is described. The manufacturing method of the connector **10** is not limited to the one described below.

[0060] The terminal body **51** and the slide portion **52** are formed by a known method. The slide portion **52** is assembled with the terminal body **51** from behind, and held at the temporary locking position. In this way, the terminal **50** is obtained.

[0061] By injection-molding the synthetic resin, the connector housing **20** and the rear holder **30** are formed. The terminals **50** are inserted into the cavities **21A** of the connector housing **20** from behind.

[0062] The rear holder **30** is assembled with the rear end part of the connector housing **20** from behind. Then, front end parts of the lock receiving portions **32** of the rear holder **30** come into contact with the temporary lock portions **24** of the connector housing **20** from behind and the lock receiving portions **32** are expanded and deformed. If the rear holder **30** is pushed further forward, the lock receiving portions **32** are restored and the lock receiving portions **32** of the rear holder **30** are resiliently locked to the temporary lock portions **24** of the connector housing **20**. In this way, the rear holder **30** is held at the temporary locking position with respect to the connector housing **20**.

[0063] By stripping the insulation coating **42** in the end part of the wire **40**, the core wire **41** is exposed over a predetermined length. The front end part of the insulation coating **42** of the wire **40** is cut off by a wire stripper or the like to form the sheath portion **43** and the body portion **44**. By sliding the sheath portion **43** forward with respect to the core wire **41**, the front end part of the core wire **41** is protected by the sheath portion **43** (see FIG. 2).

[0064] Subsequently, the wire **40** having the front end part of the core wire **41** protected by the sheath portion **43** is inserted into the wire insertion hole **33** of the rear holder **30** (see FIG. 5). In this embodiment, by properly setting the inner diameter of the wire insertion hole **33**, the core wire **41** disposed substantially in the center of the sheath portion **43** is disposed inside the insertion tube portion **55** when the sheath portion **43** disposed in the wire insertion hole **33** is viewed from behind (see FIG. 8). Further, the sheath portion **43** slides in contact with the guiding portion **34**, thereby being guided in a projecting direction (downward in FIGS. 5 and 8) of the guiding portion **34** near the front end of the wire insertion hole **33**. Since the insertion tube portion **55** is shifted in the projecting direction of the guiding portion **34** with respect to the wire insertion hole **33**, the center positions in the vertical direction of the sheath portion **43** and the insertion tube portion **55** are more easily aligned by the guiding portion **34**.

[0065] If the wire **40** is pushed further forward, the front end of the sheath portion **43** comes into contact with the contact portions **56** of the insertion tube portion **55**. In this embodiment, since the contact portions **56** are provided on the first and second wall portions **55A**, **55B** as shown in FIG. 8, the posture of the front end of the sheath portion **43** is easily stabilized by contact with the contact portions **56**. For example, the sheath portion **43** can be suppressed from being inclined from the state extending in the front-rear direction.

[0066] If the wire **40** is pushed further forward, the core wire **41** slides with respect to the sheath portion **43** and is fed into the insertion tube portion **55** (slide portion **52**) with the sheath portion **43**

kept in contact with the contact portions **56** as shown in FIG. **6**. As described above, the core wire **41** disposed substantially in the center of the sheath portion **43** is disposed inside the insertion tube portion **55** when viewed from behind (see FIG. **8**). Thus, with the sheath portion **43** held in contact with the contact portions **56**, the front end part of the core wire **41** is disposed to face the inside of the insertion tube portion **55** at a position immediately behind the insertion tube portion **55**. Therefore, the core wire **41** is more easily inserted into the insertion tube portion **55** as compared to the case where the sheath portion **43** is not provided.

[0067] If the wire **40** is pushed further forward, the front end part of the core wire **41** is inserted into the terminal body **51** and reaches the space between the upper and lower connection pieces **51C**, **51D** as shown in FIG. **7**.

[0068] By seeing the front end part of the core wire **41** or detecting this front end part by a probe or the like in the terminal window portion **51F** visually confirmable through the connector window portion **21B**, it can be confirmed that the core wire **41** has reached a predetermined position.

[0069] Subsequently, if the jig (not shown) is pushed into contact with the jig contact portion **54** from behind and the slide portion **52** is slid forward, the slide portion **52** is held at the complete locking position with respect to the terminal body **51**. With the slide portion **52** held at the complete locking position with respect to the terminal body **51**, the upper contact portion **52A** of the slide portion **52** contacts the upper connection piece **51C** of the terminal body **51** from above to press the upper connection piece **51C** downward as shown in FIG. **4**. Further, the lower contact portion **52B** of the slide portion **52** contacts the lower connection piece **51D** of the terminal body **51** from below to press the lower connection piece **51D** upward. In this way, the core wire **41** is sandwiched from upper and lower sides by the upper and lower connection pieces **51C**, **51D** and the terminal **50** and the wire **40** are electrically connected.

[0070] Subsequently, the rear holder **30** is pushed further forward with respect to the connector housing **20**. Then, the front end parts of the lock receiving portions **32** of the rear holder **30** ride on the complete lock portions **25** and are expanded and deformed. If the rear holder **30** is pushed further forward, the lock receiving portions **32** are restored and the lock receiving portions **32** of the rear holder **30** are resiliently locked to the complete lock portions **25** of the connector housing **20**. In this way, the rear holder **30** is held at the complete locking position with respect to the connector housing **20** (see FIG. **1**). In this state, the receptacle **31** of the rear holder **30** is externally fit to the rear part of the connector housing **20** and the connector housing **20** and the rear holder **30** are integrally configured. In the above way, the manufacturing of the connector **10** is completed.

Functions and Effects of Embodiment

[0071] According to the embodiment, the following functions and effects are achieved.

[0072] The connector **10** according to the embodiment is provided with the wires **40** each including the core wire **41** and the insulation coating **42** covering the outer periphery of the core wire **41**, the terminals **50** to be connected to the core wires **41**, the connector housing **20** including the terminal accommodating portion **21** for accommodating the terminals **50** inside, and the rear holder **30** to be attached to the rear part of the connector housing **20**. The rear holder **30** includes the wire insertion holes **33**, through which the wires **40** are inserted. The terminal **50** includes the insertion tube portion **55**, into which the core wire **41** is inserted from behind, and the contact portions **56** disposed in the rear end part of the insertion tube portion **55** and facing the wire insertion hole **33**. The insulation coating **42** of the wire **40** includes the sheath portion **43** and the body portion **44** separate from each other. The sheath portion **43** is disposed forward of the body portion **44** and slidable with respect to the core wire **41**.

[0073] According to this configuration, by inserting the wire **40** into the wire insertion hole **33** with the front end part of the core wire **41** protected by the sheath portion **43**, the sheath portion **43** comes into contact with the contact portions **56**. With the sheath portion **43** held in contact with the contact portions **56**, the core wire **41** immediately behind the insertion tube portion **55** is substantially in the center of the sheath portion **43** and disposed near the center of the wire insertion

hole **33**. Therefore, the core wire **41** is easily inserted into the insertion tube portion **55**.
[0074] In the embodiment, the insertion tube portion **55** includes the first wall portion **55A** and the second wall portion **55B** facing the first wall portion **55A**, the clearance **CL1** between the first and second wall portions **55A**, **55B** is set to be smaller than the outer diameter **D1** of the wire **40**, and both the first and second wall portions **55A**, **55B** include the contact portion **56**.

[0075] According to this configuration, since both the first and second wall portions **55A**, **55B** include the contact portion **56**, the state of the sheath portion **43** in contact with the contact portions **56** can be easily stabilized. In this way, the core wire **41** is more easily inserted into the insertion tube portion **55**.

[0076] In the embodiment, the rear holder **30** includes the guiding portion **34** projecting more inwardly of the wire insertion hole **33** from the inner wall of the wire insertion hole **33** toward the front.

[0077] According to this configuration, the sheath portion **43** can be guided by being caused to slide in contact with the guiding portion **34**. Thus, the core wire **41** is even more easily inserted into the insertion tube portion **55**.

[0078] In the embodiment, the manufacturing method of the connector **10** of the present disclosure is a manufacturing method for the connector **10** provided with the wires **40** each including the core wire **41** and the insulation coating **42** covering the outer periphery of the core wire **41**, the terminals **50** provided with the insertion tube portions **55**, into which the core wires **41** are inserted from behind, the connector housing **20** including the terminal accommodating portion **21**, and the rear holder **30** including the wire insertion holes **33**, and includes accommodating the terminals **50** inside the terminal accommodating portion **21**, attaching the rear holder **30** to the rear part of the connector housing **20** with the rear end parts of the insertion tube portions **55** at least partially arranged to face the wire insertion holes **33**, cutting off the front end part of the insulation coating **42** of the wire **40** to form the sheath portion **43** and the body portion **44** disposed behind the sheath portion **43**, protecting the front end part of the core wire **41** by the sheath portion **43** by sliding the sheath portion **43** forward with respect to the core wire **41**, inserting the wire **40** having the front end part of the core wire **41** protected by the sheath portion **43** into the wire insertion hole **33** of the rear holder **30** and introducing the core wire **41** into the insertion tube portion **55** while the sheath portion **43** is brought into contact with the rear end part of the insertion tube portion **55**, and connecting the core wire **41** and the terminal **50**.

[0079] Since such a manufacturing method of the connector **10** includes forming the sheath portion **43** in the insulation coating **42** and protecting the front end part of the core wire **41** by the sheath portion **43**, the sheath portion **43** comes into contact with the rear end part of the insertion tube portion **55** if the wire **40** is inserted into the wire insertion hole **33**. With the sheath portion **43** held in contact with the rear end part of the insertion tube portion **55**, the core wire **41** immediately behind the insertion tube portion **55** is located substantially in the center of the sheath portion **43** and disposed near the center of the wire insertion hole **33**. Therefore, the core wire **41** is easily inserted into the insertion tube portion **55**.

Other Embodiments

[0080] (1) Although both the first and second wall portions **55A**, **55B** of the insertion tube portion **55** include the contact portion **56** in the above embodiment, there is no limitation to this and at least a part of a rear end part of an insertion tube portion may include a contact portion. [0081] (2) Although the rear holder **30** is provided with the guiding portions **34** in the above embodiment, there is no limitation to this and the guiding portions may be omitted.

LIST OF REFERENCE NUMERALS

[0082] **10**: connector [0083] **20** connector housing [0084] **21**: terminal accommodating portion [0085] **21A**: cavity [0086] **21B**: connector window portion [0087] **22** separation wall [0088] **23**: partition wall [0089] **24**: temporary lock portion [0090] **25** complete lock portion [0091] **26**: lance hole [0092] **30**: rear holder [0093] **31**: receptacle [0094] **32**: lock receiving portion [0095] **33**: wire

insertion hole [0096] **34**: guiding portion [0097] **35**: separation wall receiving portion [0098] **36**: step [0099] **40**: wire [0100] **41**: core wire [0101] **42**: insulation coating [0102] **43**: sheath portion [0103] **44**: body portion [0104] **50**: terminal [0105] **51**: terminal body [0106] **51A**: connecting tube portion [0107] **51B**: base portion [0108] **51C**: upper connection piece [0109] **51D**: lower connection piece [0110] **51E**: resilient contact piece [0111] **51F**: terminal window portion [0112] **52**: slide portion [0113] **52A**: upper contact portion [0114] **52B**: lower contact portion [0115] **53**: locking lance [0116] **54**: jig contact portion [0117] **55**: insertion tube portion [0118] **55A**: first wall portion [0119] **55B**: second wall portion [0120] **56**: contact portion [0121] **CL1**: clearance between first and second wall portions [0122] **D1**: outer diameter of wire [0123] **D2**: diameter of core wire

Claims

1. A connector, comprising: a wire including a core wire and an insulation coating covering an outer periphery of the core wire; a terminal to be connected to the core wire; a connector housing including a terminal accommodating portion for accommodating the terminal inside; and a rear holder to be attached to a rear part of the connector housing, the rear holder including a wire insertion hole, the wire being inserted through the wire insertion hole, the terminal including an insertion tube portion, the core wire being inserted into the insertion tube portion from behind, and a contact portion disposed in a rear end part of the insertion tube portion and facing the wire insertion hole, the insulation coating of the wire including a sheath portion and a body portion separate from each other, and the sheath portion being disposed forward of the body portion and slidable with respect to the core wire.
 2. The connector of claim 1, wherein: the insertion tube portion includes a first wall portion and a second wall portion facing the first wall portion, a clearance between the first and second wall portions is set to be smaller than an outer diameter of the wire, and both the first and second wall portions include the contact portion.
 3. The connector of claim 1, wherein the rear holder includes a guiding portion projecting more inwardly of the wire insertion hole from an inner wall of the wire insertion hole toward front.
 4. A connector manufacturing method for a connector with a wire including a core wire and an insulation coating covering an outer periphery of the core wire, a terminal including an insertion tube portion, the core wire being inserted into the insertion tube portion from behind, a connector housing including a terminal accommodating portion, and a rear holder including a wire insertion hole, comprising: accommodating the terminal inside the terminal accommodating portion; attaching the rear holder to a rear part of the connector housing with a rear end part of the insertion tube portion at least partially arranged to face the wire insertion hole; forming a sheath portion and a body portion disposed rearward of the sheath portion by cutting off a front end part of the insulation coating of the wire; protecting a front end part of the core wire by the sheath portion by sliding the sheath portion forward with respect to the core wire; inserting the wire having the front end part of the core wire protected by the sheath portion into the wire insertion hole of the rear holder and introducing the core wire into the insertion tube portion while bringing the sheath portion into contact with the rear end part of the insertion tube portion; and connecting the core wire and the terminal.
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